Aalto University
Department of Mathematics and Systems Analysis
MS-E1200 - Lie groups and Lie algebras

Problem set 3
2025-2026/IV
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Exercise session: Thu 12.3. at 14-16 Hand-in due: Mon 16.3. at 12:00

In the weekly exercise sheets there are two types of problems: problems whose solutions are to be presented in the exercise session, and hand-in problems for which written solutions are to be returned.

Exercise 1. The set $\mathbb{C}^{n\times n}$ of $n \times n$-matrices is naturally identified with the space of linear maps $\mathbb{C}^n \to \mathbb{C}^n$, and is equipped with the operator norm: the norm of $X \in \mathbb{C}^{n\times n}$ is

$$\|X\|_{\mathrm{op}} = \sup\left\{ \|Xv\| \;\middle|\; v \in \mathbb{C}^n,\ \|v\| = 1 \right\},$$

where $\|\cdot\|$ denotes the usual norm on $\mathbb{C}^n$.

(a) If $X, Y \in \mathbb{C}^{n\times n}$ are two matrices, show that $\|XY\|_{\mathrm{op}} \leq \|X\|_{\mathrm{op}}\,\|Y\|_{\mathrm{op}}$.

(b) For $X \in \mathbb{C}^{n\times n}$, show that the following series is convergent in $\mathbb{C}^{n\times n}$:

$$\exp(X) = \sum_{n=0}^{\infty} \frac{1}{n!} X^n.$$

(c) For $t, s \in \mathbb{R}$ and $X \in \mathbb{C}^{n\times n}$, show that $\exp(sX)\exp(tX) = \exp\big((s+t)X\big)$.

(d) For any matrix $Z \in \mathbb{C}^{n\times n}$ and an invertible matrix $M \in \mathbb{C}^{n\times n}$, denote by $\mathrm{Ad}_M(Z) = MZM^{-1}$ the conjugation of Z by M. Show that

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mathrm{Ad}_{\exp(tX)}(Z) = XZ - ZX.$$

Exercise 2. Let G be a connected Lie group, and $U \subset \mathrm{G}$ an open neighborhood of the neutral element $e \in \mathrm{G}$. Show that the elements in U generate the entire group G (i.e., that the smallest subgroup of G containing U is the entire group).

Hint: Recall that in a connected topological space, the empty set and the whole space are the only subsets which are both open and closed (i.e., complement is open).

Exercise 3. In this exercise we consider the symplectic group and its Lie algebra. The symplectic group is the group of linear transformations which preserves a symplectic form (just like the orthogonal group is the group of linear transformations which preserve an inner product).

Let $J \in \mathbb{R}^{2n\times 2n}$ be an antisymmetric matrix such that $J^2 = -\mathbb{I}$. Define a bilinear form $\boldsymbol{\omega}$ on $\mathbb{R}^{2n}$ by $\boldsymbol{\omega}(v,w) = v^\top J w$. Let $\mathrm{Sp}_{2n}(\mathbb{R})$ be the set of those $M \in \mathbb{R}^{2n\times 2n}$ which preserve the form $\boldsymbol{\omega}$ in the sense that for all $v, w \in \mathbb{R}^{2n}$ we have $\boldsymbol{\omega}(v,w) = \boldsymbol{\omega}(Mv, Mw)$.

(a) Show that we have $\mathrm{Sp}_{2n}(\mathbb{R}) = \left\{ M \in \mathbb{R}^{2n\times 2n} \,\middle|\, M^\top J M = J \right\}$, and show that $\mathrm{Sp}_{2n}(\mathbb{R}) \subset \mathrm{GL}_{2n}(\mathbb{R})$ is a closed subgroup[1].

(b) Find a linear condition (on X), which is necessary and sufficient for a matrix $X \in \mathbb{R}^{2n\times 2n}$ to satisfy $\exp(sX) \in \mathrm{Sp}_{2n}(\mathbb{R})$ for all $s \in \mathbb{R}$. Denote the set of such X by $\mathfrak{sp}_{2n}(\mathbb{R})$. Show directly using the linear condition you found that whenever $X, Y \in \mathfrak{sp}_{2n}(\mathbb{R})$, then also $[X,Y] = XY - YX \in \mathfrak{sp}_{2n}(\mathbb{R})$.

✍ **Hand-in 4.** For $X \in \mathbb{C}^{n\times n}$, let $e^X = \sum_{k=0}^{\infty} \frac{1}{k!} X^k$ and when $\|X\|_{\mathrm{op}} < 1$, also $\log(\mathbb{I} + X) = -\sum_{k=1}^{\infty} \frac{1}{k}(-X)^k$.

(a) Let $X \in \mathbb{C}^{n\times n}$, and define

$$f(t) = \det(e^{tX}).$$

for all $t \in \mathbb{R}$. Show that f satisfies the differential equation $f'(t) = c\, f(t)$, where $c = \mathrm{tr}(X)$, and the initial condition $f(0) = 1$. Deduce that

$$\det(e^X) = e^{\mathrm{tr}(X)}.$$

(b) Prove the Baker-Campbell-Hausdorff formula up to third order: for $X, Y \in \mathbb{C}^{n\times n}$ with $\|X\|_{\mathrm{op}}, \|Y\|_{\mathrm{op}}$ sufficiently small, we have

$$\log\left(e^X e^Y\right) = X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}[X,[X,Y]] + \frac{1}{12}[Y,[Y,X]] + \cdots,$$

where the omitted terms $\cdots$ are $\mathcal{O}\big(\max(\|X\|_{\mathrm{op}}^4, \|Y\|_{\mathrm{op}}^4)\big)$.

✍ **Hand-in 5.** Show that the set $\mathrm{SO}_n = \left\{ M \in \mathbb{R}^{n\times n} \,\middle|\, M^\top M = \mathbb{I},\ \det(M) = 1 \right\}$ is path-connected[2], i.e., that for any $M \in \mathrm{SO}_n$, there exists a continuous path $\gamma\colon [0,1] \to \mathrm{SO}_n$ such that $\gamma(0) = M$ and $\gamma(1) = \mathbb{I}$.

Hint: There are many possible ways, but one concrete approach is to implement the following idea. Noting that elements of SO_n can be identified with ordered and oriented orthonormal bases $v_1, \ldots, v_n$ in $\mathbb{R}^n$, it is sufficient to continuously deform a given orthonormal basis to the standard basis. To do this, it is convenient to first show that the first basis vector v_1 can be continuously rotated to the first standard basis vector. Then inductively in the dimension n, one can show that in the $n-1$-dimensional orthogonal complement to the first standard basis vector, one can deform the remaining basis vectors.

[1] *Closed subgroup* means a subgroup which is a closed set, topologically. A subgroup is by definition stable under multiplication ("closed under multiplication").

[2] Generally in topology, *path-connected* is a stronger notion than *connected* (path-conn. $\Rightarrow$ conn.). For manifolds, and Lie groups in particular, the two notions are nevertheless logically equivalent.